



Yasser Ahmed Mogahed



Supervised by:

Ali Zein Elabdeen Mohamed Abdelnaby

1. Project Overview:

The Maven Market project is a business intelligence solution built using Power BI to analyze sales, customers, products, and operational performance of a multi-branch grocery retail company.

✓ Business Problem:

Retail managers need clear visibility into:

- Overall revenue and profitability
- Store and regional performance
- Product contribution and margins
- Customer purchasing patterns
- Sales trends over time

Without a centralized reporting system, decision-makers struggle to identify performance gaps and optimize strategies.

✓ The objective of the project is to:

- Monitor sales performance across regions and stores
- Analyze product profitability
- Evaluating customer behavior
- Track revenue, cost, and profit trends
- Support data-driven decision-making

2. Data Model & Architecture:

The model consists of the following tables:

✓ **Fact Tables:**

1. Transactions_1997-1998
2. Returns_1997-1998

✓ **Dimension Tables:**

- Products
- Customers
- Stores
- Regions
- Calendar

✓ **Data Modeling Approach**

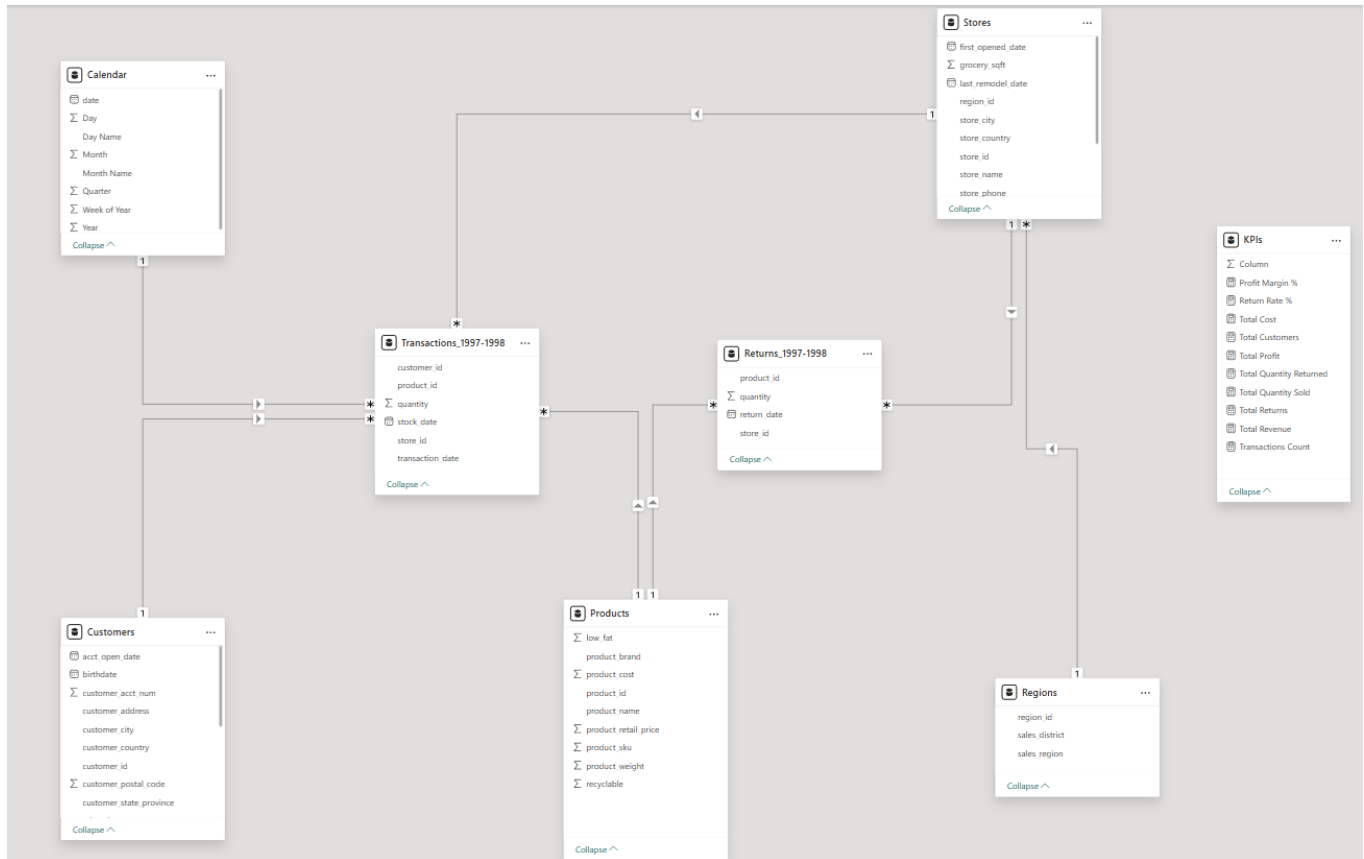
The model follows a dimensional structure where:

- Transactions_1997-1998 acts as the primary sales fact table
- Returns_1997-1998 acts as a secondary fact table
- Products, Customers, Stores, Regions, and Calendar serve as dimension tables

Relationships are primarily One-to-Many from dimension tables to fact tables. The Calendar table is used to enable time-based analysis such as monthly and yearly performance tracking.

This structure supports efficient filtering and dynamic aggregation of KPIs.

✓ Data Model Diagram:



3. Key Performance Indicators (KPIs):

The dashboard includes the following core KPIs:

✓ **Financial KPIs**

- **Total Revenue** – Total sales value generated.
- **Total Cost** – Total cost of products sold.
- **Total Profit** – Revenue minus cost.
- **Profit Margin %** – Profit as a percentage of revenue.

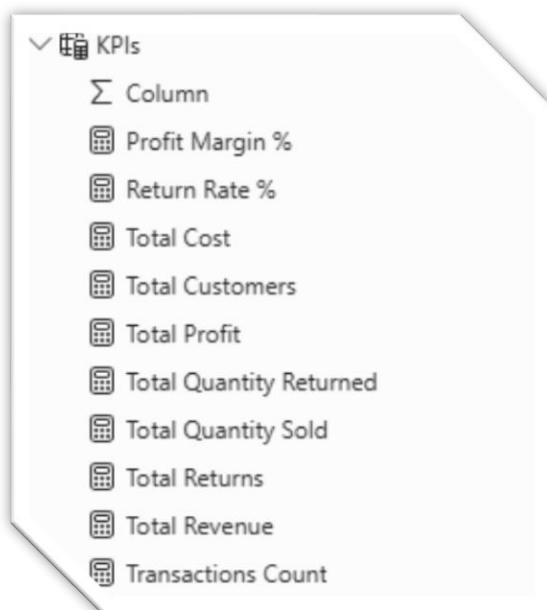
✓ **Sales KPIs**

- **Transactions Count** – Total number of sales transactions.
- **Total Quantity Sold** – Total units sold.
- **Total Customers** – Number of unique customers.

✓ **Returns KPIs**

- **Total Returns** – Number of return transactions.
- **Total Quantity Returned** – Units returned.
- **Return Rate %** – Returned quantity relative to sold quantity.

These KPIs allow management to monitor both revenue growth and operational efficiency.



4. Report Pages & Analytical Insights:

✓ Executive Overview Page:

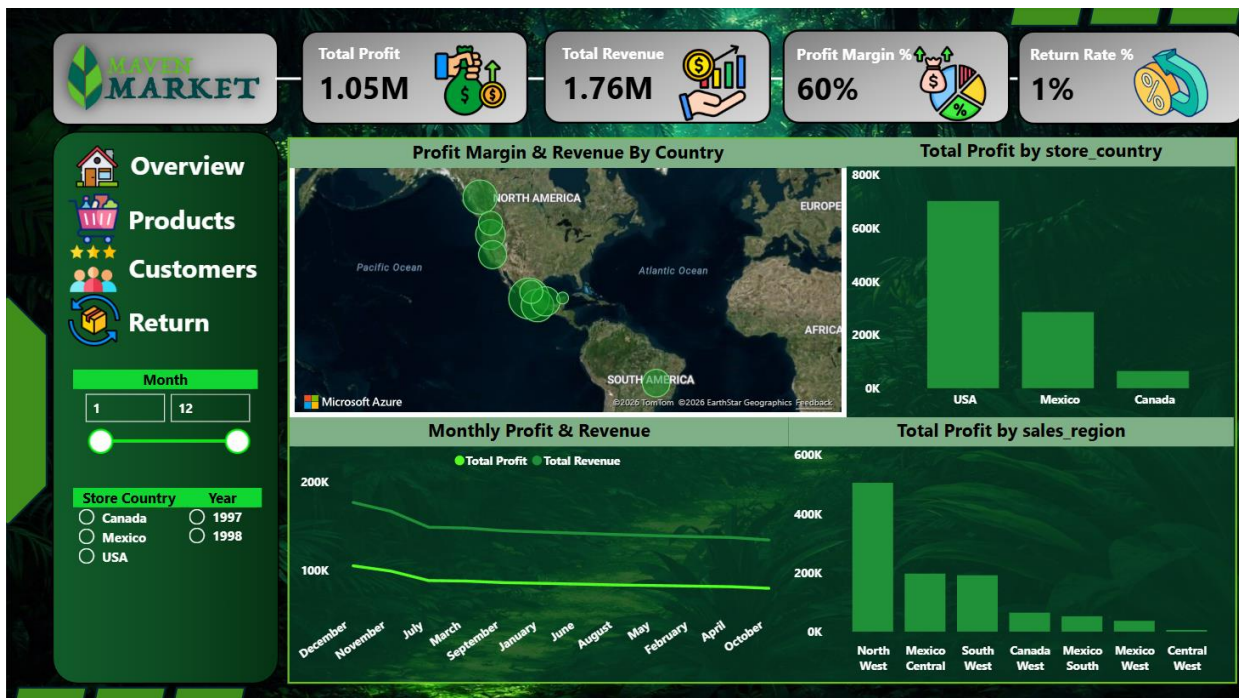
This page provides a high-level summary of business performance using KPI cards and trend visuals.

Key metrics displayed:

- Total Revenue
- Total Profit
- Profit Margin %
- Return Rate %
- Transactions Count

Insights derived:

- Overall financial health of the company
- Impact of returns on profitability
- Sales trends over time



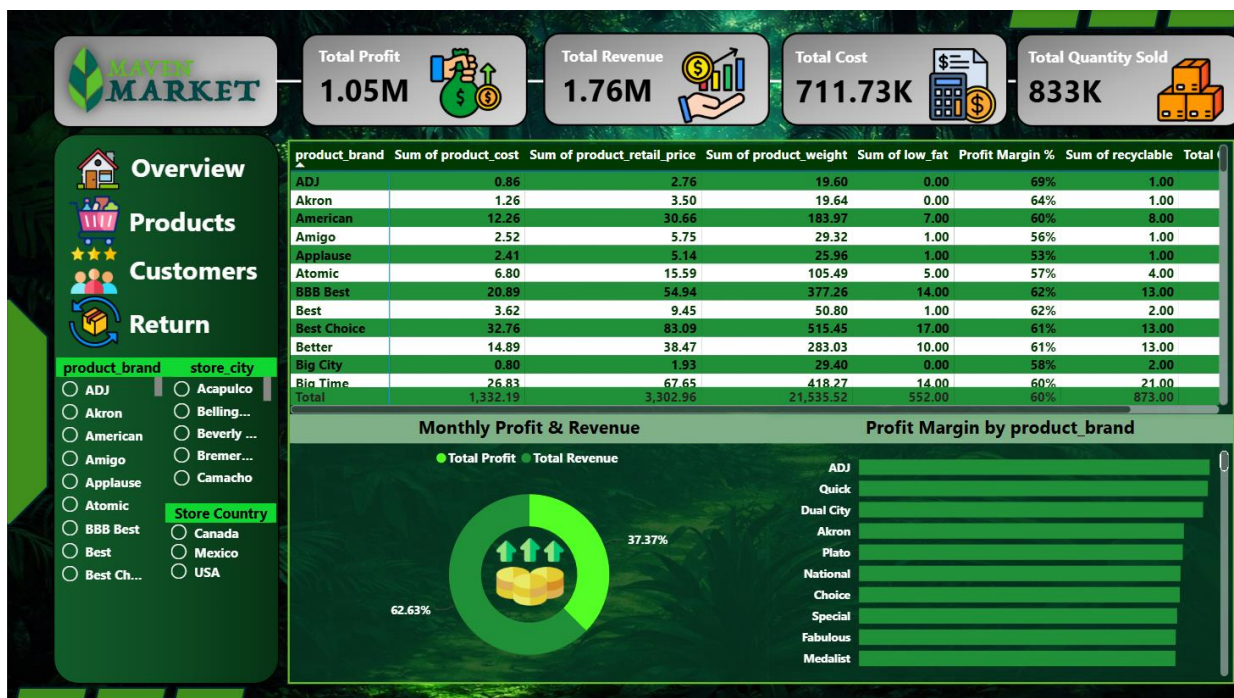
✓ Products Page

This page evaluates product-level efficiency and profitability.

Insights:

- High-revenue but low-margin products
- Most profitable product categories
- Products with high return rates
- Impact of cost structure on overall profit

This analysis helps management optimize pricing and inventory strategies.



✓ Customer Insights Page

This page examines customer contribution and behavior.

Metrics analyzed:

- Revenue by Customer
- Total Customers
- Customer distribution by region

Insights:

- Customer concentration impact
- Identification of high-value customers
- Geographic customer patterns



✓ Returns Page:

This page provides a high-level overview of the company's return performance, revenue, and profitability across countries, store types, and product brands. It combines KPI cards, maps, and trend visuals to summarize business health and operational insights.

Key Metrics Displayed:

- **Return Rate:** 1%
- **Total Quantity Returned:** 8,000 units
- **Total Returns:** 7,000 transactions
- **Total Transactions:** 270,000
- **Monthly Profit & Revenue:** Shown as a downward trend from 13% to ~10% profit margin
- **Return Rate by Brand:** Top brands with highest returns include King (1.8%) and Genteel (1.5%)

Insights Derived:

- Overall returns are low (1%), indicating strong product performance and customer satisfaction.
- Profit margin shows a downward trend, partially affected by returns in peak months.
- Store type and brand analysis highlights where returns are concentrated, informing inventory and quality management.
- Geographic distribution provides a global view of sales performance and potential market focus areas.



5. Conclusion:

- The Maven Market Power BI project successfully transforms transactional sales and returns data into a structured Business Intelligence solution that supports strategic decision-making.
- By integrating financial, operational, and customer metrics into a centralized dashboard, the solution provides clear visibility into business performance. The inclusion of both sales and return analysis enhances profitability monitoring and operational control.
- This project demonstrates the practical application of dimensional modeling, KPI design, DAX calculations, and interactive dashboard development. It highlights the strategic value of Business Intelligence in improving retail performance, optimizing cost management, and supporting data-driven decision-making.
- Overall, the solution serves as a scalable and effective analytical framework capable of supporting executive-level reporting and operational optimization.